

7.BINARY SEARCH:

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[100], n, key, low, high, mid;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter sorted elements:\n");
```

```
    for (int i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    printf("Enter element to search: ");
```

```
    scanf("%d", &key);
```

```
    low = 0;
```

```
    high = n - 1;
```

```
    while (low <= high) {
```

```
        mid = (low + high) / 2;
```

```
        if (a[mid] == key) {
```

```
            printf("Element found at position %d", mid + 1);
```

```
            return 0;
```

```
        }
```

```
        else if (key < a[mid])
```

```
            high = mid - 1;
```

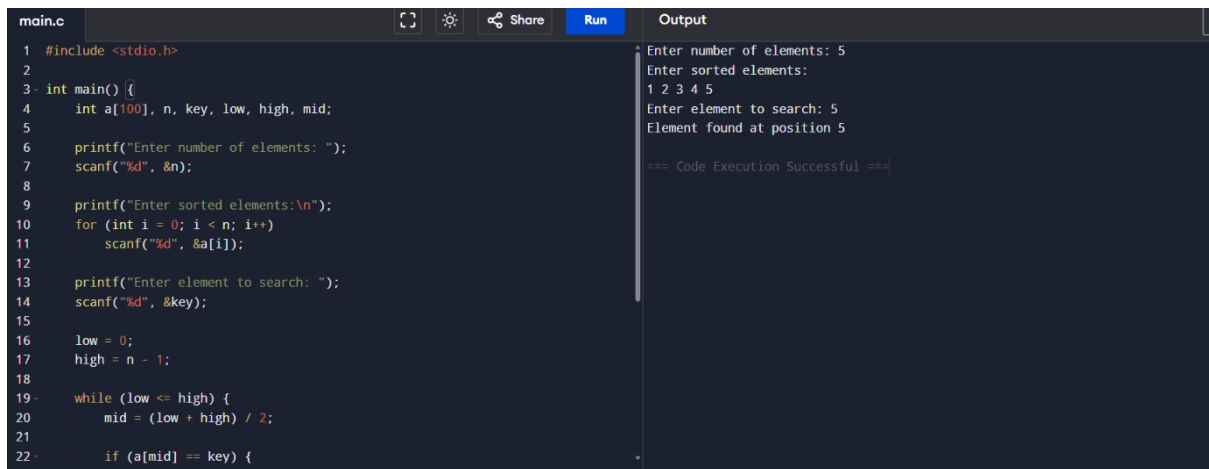
```
        else
```

```
        low = mid + 1;
    }
}
```

```
printf("Element not found");
```

```
return 0;
}
```

OUTPUT:



The screenshot shows a C code editor with a file named 'main.c'. The code implements a binary search algorithm. It prompts the user to enter the number of elements (5), the sorted elements (1 2 3 4 5), and the element to search (5). The output shows that the element was found at position 5. The code execution was successful.

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key, low, high, mid;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     printf("Enter sorted elements:\n");
10    for (int i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    printf("Enter element to search: ");
14    scanf("%d", &key);
15
16    low = 0;
17    high = n - 1;
18
19    while (low <= high) {
20        mid = (low + high) / 2;
21
22        if (a[mid] == key) {
```

Output:

```
Enter number of elements: 5
Enter sorted elements:
1 2 3 4 5
Enter element to search: 5
Element found at position 5

=== Code Execution Successful ===
```